Math 140H — Honors Calculus I: Lecture Notes

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2026-08-25

Table of contents

Welcome!

These notes accompany Math 140H, Honors Calculus I, Fall 2026. Each lecture here corresponds to one class session and covers more or less the material presented in class, in the same order, so that it can serve as a compact study reference. HTML "hardcopies" of the Jupyter notebook activities from class are also included. Lecture notes will generally be available immediately **after** the corresponding class session.

State College, Fall 2026

Part I

Real Numbers

1 The Rational Numbers and the Number Line

> Companion reading
>
> This lecture corresponds to C&J §1.1a, together with the neighborhood definition from §1.1d — covered a bit early here since we will use it constantly for the rest of the semester.

Calculus is built on the real numbers, so before we can talk about limits, derivatives, or integrals we need to understand exactly what a real number *is* — and, just as importantly, why the rational numbers alone are not enough. This lecture makes that precise: we recall how the natural numbers extend to the rationals, set up the number line, and then prove that a length as ordinary as the hypotenuse of a simple right triangle cannot be a rational number at all.

1.1 From Counting to Measuring

There are two major uses for numbers: **counting** and **measuring**. The natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ are the tool for counting — abstract symbols for how many elements are in a collection, stripped of any reference to what is actually being counted. They come equipped with addition and multiplication satisfying a short list of familiar rules.

Remark 1.1. For all natural numbers a, b, c:

- $a + b = b + a$ and $ab = ba$ (commutative laws)
- $a + (b + c) = (a + b) + c$ and $a(bc) = (ab)c$ (associative laws)
- $a(b + c) = ab + ac$ (distributive law)
- $a + c = b + c \;\Rightarrow\; a = b$ (cancellation law)

These are the *rules* of arithmetic, not its *content* — everything else in this course is built on top of them, and we will not dwell on them further.

Natural numbers can be added and multiplied, but not always subtracted or divided. There is no natural number a with

$$5 + a = 2,$$

so subtraction is not always possible in $\mathbb{N}$. Extending $\mathbb{N}$ to the **integers** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ fixes this: now $5 + (-3) = 2$.

In $\mathbb{Z}$, we still cannot always divide: there is no integer $b \in \mathbb{Z}$ with

$$5 \cdot b = 2.$$

Extending $\mathbb{Z}$ to the **rational numbers** fixes this in turn.

Definition 1.1.
A **rational number** is a number that can be written as p/q where $p, q \in \mathbb{Z}$ and $q \neq 0$. Every rational number has a unique representation p/q with $q > 0$ and $\gcd(p, q) = 1$ (no common factor larger than 1); we call this representation **in lowest terms**.

Within $\mathbb{Q}$, all four rational operations — addition, subtraction, multiplication, and division (except by 0) — can be carried out freely, and they obey exactly the same laws as in Remark 1.1. In this precise sense, $\mathbb{Q}$ is the smallest extension of the natural numbers in which counting-type arithmetic becomes measuring-type arithmetic.

1.2 The Number Line

To turn numbers into geometry, fix a line L, a point O on it called the **origin**, and a second point I to the right of O; the segment OI serves as our unit scale for measuring — a "prototype meter." The direction from O to I is called **positive**.

Every point on L is now uniquely determined by two pieces of information: its **distance** from O, and its **direction** (left or right) from O. For example, the point at distance $\frac{1}{2}$ in the positive direction represents the rational number $\frac{1}{2}$; the point at distance 2 in the positive direction represents 2.

Definition 1.2.
The distance of a number x from the origin is written $|x|$ and called the **absolute value** of x:

$$|x| \;=\; \begin{cases} x & x \geq 0 \\ -x & x < 0. \end{cases}$$

Note $|x| \geq 0$ always, with equality only when $x = 0$.

Every rational point can in fact be *constructed*, not just described: to locate p/q on L with ruler and compass, construct a subdivision of the segment OI into q equal parts, then extend the part closest to O, p times. We will identify each rational number with its representation on L.

1.3 Distances and Intervals

For $x, y \in L$, the number $|x-y|$ is the **distance between x and y**. Observe: if $x, y \in \mathbb{Q}$, then $|x-y| \in \mathbb{Q}$ as well — distances between rational points are themselves rational.

Definition 1.3.
For $a < b$, the **interval** between a and b comes in four flavors:

- open: $(a, b) := \{x : a < x < b\}$ (both endpoints excluded)
- closed: $[a, b] := \{x : a \leq x \leq b\}$ (both endpoints included)
- half-open: $[a, b) := \{x : a \leq x < b\}$ and $(a, b] := \{x : a < x \leq b\}$

Definition 1.4.
Given a number a and $\varepsilon > 0$, the **ε-neighborhood of a** is the open interval

$$\{y : |y-a| < \varepsilon\} = (a-\varepsilon, a+\varepsilon),$$

i.e. the set of points less than ε away from a.

We will meet ε-neighborhoods constantly once we get to limits — keep this definition close by.

1.3.1 Density

Proposition 1.1.
Between any two distinct rational numbers $a < b$ there is another rational number c with $a < c < b$. Consequently, between a and b there are infinitely many rational numbers.

Proof. Let $c = \dfrac{a+b}{2}$. Since $a, b \in \mathbb{Q}$ and $\mathbb{Q}$ is closed under addition and division by 2, $c \in \mathbb{Q}$; and since $a < b$, averaging gives $a < c < b$.

Now repeat the argument on the pair a, c to get a rational number strictly between a and c, hence strictly between a and b. Doing this again and again produces infinitely many rational numbers between a and b. $\square$

We say the rationals are **dense** on the number line: no matter how closely you look, you will always find another one.

1.4 Incommensurable Lengths

Given how constructible and how dense $\mathbb{Q}$ is, a natural question arises: is there anything "left in between" the rational numbers? Are there any *gaps*? Clearly, such a gap could not itself be an interval — density rules that out, since any interval of positive length contains infinitely many rational points. If a gap exists at all, it can only be a single missing point.

Remark 1.2. The discovery that not every length is a rational multiple of a chosen unit is due to the school of Pythagoras, in the fifth or sixth century BCE. Consider a right triangle with both legs of length 1; by the Pythagorean theorem, its hypotenuse a satisfies

$$1^2 + 1^2 = a^2 = 2,$$

and a can certainly be constructed with ruler and compass. The theorem below shows this length cannot be written as p/q for integers p, q — the hypotenuse and the legs of this triangle are, in the language of the ancients, **incommensurable**: there is no common unit of which both are integer multiples.

Theorem 1.1.
There is no rational number x with $x^2 = 2$. Equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Proof. Suppose, toward a contradiction, that $\sqrt{2} = p/q$ for integers p, q with $q \neq 0$ and $\gcd(p, q) = 1$ (i.e. p/q is in lowest terms).

1. Square both sides and clear the denominator: $p^2 = 2q^2$.
2. Conclude that p^2 is even — if p were odd, p^2 would be odd, but $p^2 = 2q^2$ is even. Write $p = 2r$ for some integer r.
3. Substitute $p = 2r$ into the equation from Step 1 and simplify to get an equation for q^2 in terms of r: $(2r)^2 = 2q^2 \;\Rightarrow\; 4r^2 = 2q^2 \;\Rightarrow\; q^2 = 2r^2$.
4. Conclude that q is also even, by the same reasoning as in Step 2.
5. Steps 2 and 4 show that 2 divides both p and q. This contradicts $\gcd(p, q) = 1$, which we assumed when we wrote p/q in lowest terms.
6. Conclude: $\sqrt{2}$ is irrational — no integers p, q satisfy $\sqrt{2} = p/q$, so $\sqrt{2} \notin \mathbb{Q}$.

□

Exercise 1.1. Adapt the proof of Theorem 1.1 to show that $\sqrt{3}$ is irrational. Then try to see where the argument would break if you attempted the same proof for $\sqrt{4}$ — since $\sqrt{4} = 2$ is certainly rational, something in the argument *must* fail. Where, exactly?

1.5 Summary

- The rationals $\mathbb{Q}$ satisfy all the usual laws of arithmetic (Remark 1.1) and are **dense**: between any two rationals lies another (Proposition 1.1).
- Distances $|x-y|$, intervals, and ε-neighborhoods (Definition 1.3, Definition 1.4) give us the basic vocabulary for talking about "closeness" on the number line — vocabulary we will lean on heavily once limits arrive.
- Density notwithstanding, $\mathbb{Q}$ does not exhaust the number line: Theorem 1.1 exhibits an ordinary, constructible length that is not a rational number.
- The rationals therefore leave *gaps* in the line. Lecture 2 closes these gaps rigorously, using **nested intervals** to construct the real numbers.